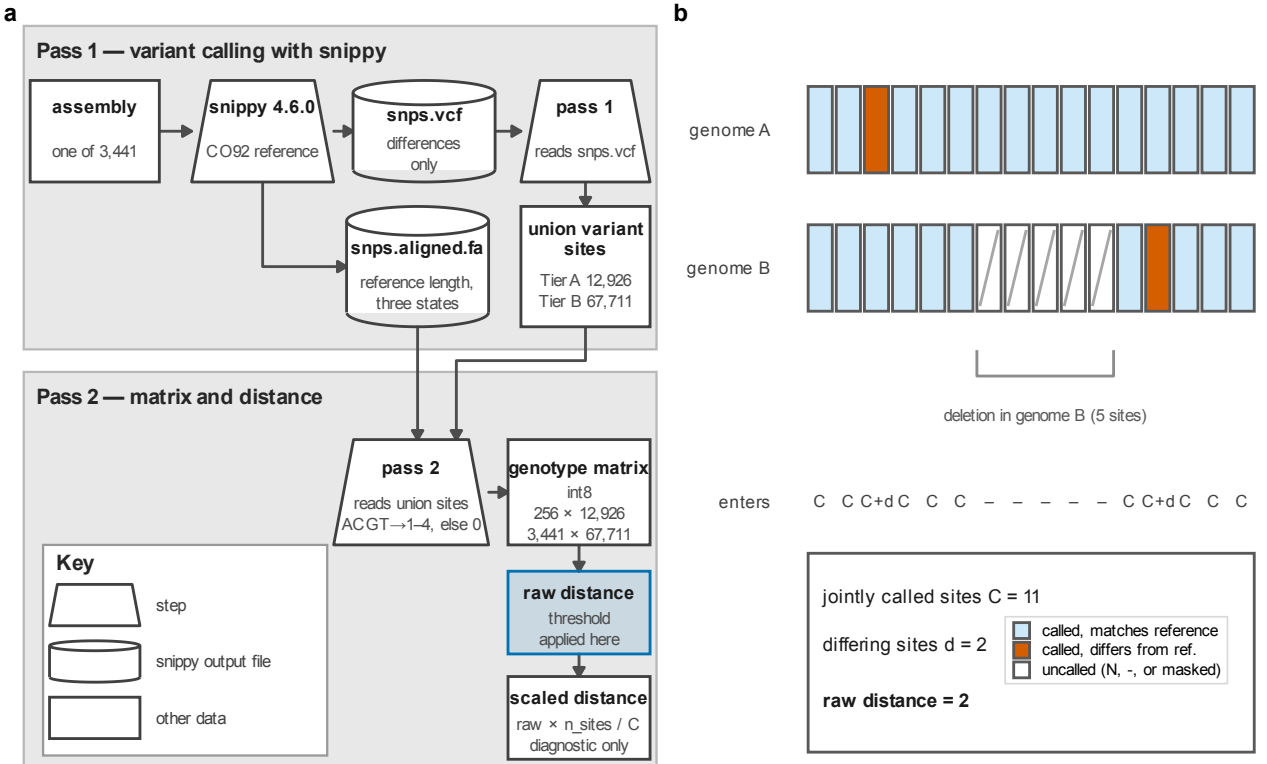

a
Pass 1 — variant calling with snippy
assembly
one of 3,441
snippy 4.6.0
CO92 reference
snps.vcf
differences only
pass 1
reads snps.vcf
snps.aligned.fa
reference length, three states
union variant sites
Tier A 12,926
Tier B 67,711
Pass 2 — matrix and distance
pass 2
reads union sites
ACGT→1–4, else 0
genotype matrix
int8
256 × 12,926
3,441 × 67,711
raw distance
threshold applied here
scaled distance
raw × n_sites / C
diagnostic only
Key
step
snippy output file
other data
b
genome A
genome B
deletion in genome B (5 sites)
enters
C C C+d C C C – – – – – C C+d C C C
jointly called sites C = 11
differing sites d = 2
called, matches reference
called, differs from ref.
uncalled (N, -, or masked)
raw distance = 2